

AIMLZG546 - Software Engineering for Machine Learning

Assignment I

Overview

This assignment assesses your ability to apply software engineering principles to a machine learning application — from problem formulation and requirements engineering through to system architecture and implementation. You will work in groups to select a real-world ML application domain, model it using GR4ML notation, and design and implement a working system architecture.

General Instructions

- It is a Group Assignment. Please follow the naming convention as <Group no>.ipynb.
- Inside each report and implementation notebooks, you are required to mention your name, Group details.

Group Details

Group No: <Fill Group No>

Group Member Names with Contribution

Sl. No	BITS ID	Name	Contribution of team member (Qualitative)	Percentage Contribution (Out of 100) Quantitative
1				
2				
3				
4				

- Weightage - 10 marks
- Assignment 1 Submission Date – 09th October 2026, 23:49.
- Submissions after the due date will result in deduction of marks.

Note

1. No extension of deadline will be given under any circumstances. Manage your time and complete the assignment within the deadline.
2. Since Assignment-I is part of EC-1 component, there is NO MAKEUP for this component.
3. Any questions or clarifications regarding the assignment can be emailed to "omshree.b@wilp.bits-pilani.ac.in".

Project Details

Objective 1: Requirements Formulation [5 Marks]

1. Select a domain and define an appropriate problem statement for an ML-based application of your choice.
2. Formulate the requirement specifications and establish measurable goals for the proposed ML-based application using GR4ML concepts.
3. Using GR4ML notations, develop the following views for the application:
 - Business View
 - Analytics Design View
 - Data Preparation View
4. Identify the top three quality requirements for the application and provide justification for their selection.

Objective 2: System Architecture [5 Marks]

5. Draw a system architecture diagram for the selected application, clearly illustrating both ML and non-ML components. You may use any suitable tool of your choice to create the diagram.
6. Select and apply any two relevant architectural patterns suitable for the application.
7. Implement the selected architectural patterns using appropriate technologies and tools of your choice.

Submission Guidelines

Prepare a Word/ PDF document containing the project details (screenshots of the application are required, along with explanation) and code. Highlight clearly the contribution of each group member in executing the assignment. Upload the file as <groupid.docx> or <groupid.pdf> into the Taxila portal.

All the Best!